Marked Mobius orbits with quadratic collision recovery

Point action -> six scalar images

T([z]G)=[1/(1-z)]G from the transverse norm; S(W)=G-W

Canonical point -> retained transporter

Least encoded image and normalized H; generic canonicalization charges two changing-target norms

Known affine move -> closed matrix tracker

M <- H A M; f_M(k)=(ak+b)/(ck+d). Rational degree stays one.

Nontrivial collision -> checked roots

Cross-multiply M1 and M2; degree <= 2. Reject proportional matrices; verify roots against Q.

Construction, 2026-11-08, T-1939. The Round 60 norm primitive now supports an implemented orbit-representative method with explicit transporter matrices and scalar recovery. Across five frozen or amended small-curve panels, all 63 targets recover: nine through nontrivial collisions, nineteen through order-three stabilizers and thirty-five through known-value orbit equations. Five tautological attempts are retained and restarted under the frozen rule. This is demonstrated on the declared cases, not a mixing, runtime or reliability claim. All scientific statements remain unsigned producer derivations or exact checks. [pcc:unverified]

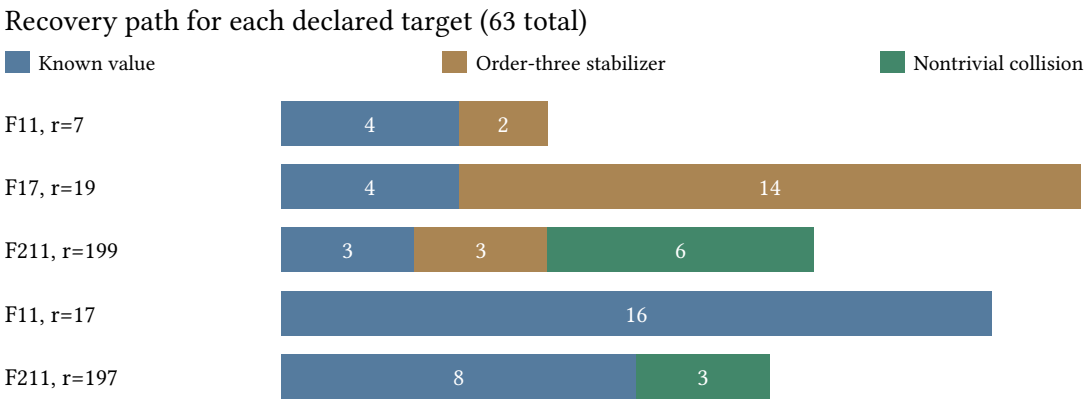
Marked action and representatives. For W=[z]G the candidate evaluates T(W)=[1/(1-z)]G with the complete transverse norm. S(W)=G-W encodes 1-z. The six S 3 images are z,Tz,T^2z and their complements. Choose the lexicographically least point encoding and retain its normalized 2x 2 transporter. A generic canonicalization needs two norm targets; each target-dependent graph constructor is charged. If T fixes a point, the second call is reused. Public special values and the projective infinity tag are handled separately; finite scalar 0 is the group identity. The reference backend uses known scalar labels and is explicitly separate from the candidate. [pcc:unverified]

Closed scalar tracking. A walk state keeps f_M(k)=(a*k+b)/(c*k+d). Both canonicalization and known affine moves update M by left multiplication, so the rational degree stays 1. The frozen moves are z+1,2z+3,3z+5,5z+7, chosen from an FNV 1a hash of the encoded state and restart index. Adding the original unknown target to a generic Mobius state would raise the degree and is excluded. The walk has four declared restart offsets and a cap of r+2 stages per attempt. All inputs, moves, failures and stopping rules were filed before affected outputs. [pcc:unverified]

Scalar collision equation. Equal finite encoded points from M 1,M 2 yield(a 1*c 2-a 2*c 1)k^2+(a 1*d 2+b 1*c 2-a 2*d 1-b 2*c 1)k+(b 1*d 2-b 2*d 1)=0. Projectively proportional matrices produce the zero polynomial and are rejected. A nontrivial collision gives a linear or quadratic equation; Tonelli-Shanks and public point verification select the scalar. Known-value and order-three stabilizer orbits also give explicit equations, recorded separately in every trace. orbit-proof.md contains the algebra, exception handling and cost argument. [pcc:unverified]

Complete orbit checks. The program qualifies all 444 projective scalar states across group orders 7,19,199,17,197, checking 2664 transformed point/matrix pairs. It retains every representative and transporter. There are 81 orbits in total: three size 2,ten size 3 and 68 size 6. The free-orbit rearrangement check verifies 816 identities H(Sz)*S=H(z) and H(Tz)*T=H(z). These are deliberately tautological comparisons: an orbit identity alone does not give a scalar. Genuine walk collisions require nonproportional accumulated transporters. [pcc:unverified]

Recovery paths, not an aggregate oracle win. The original r 7 andr 19 panels recover largely through exceptional orbits; the r 199 panel contributes six genuine collisions. Exact P-192 arithmetic shows its scalar field is 2mod 3, with no order-three stabilizer roots. Amendment 1 therefore adds the r 17 twist control; amendment 2 adds a larger matched control selected solely by a frozen point-count rule. It chooses E/F 211:y^2=x^3-3x+5 of prime order 197 after retaining the earlier rejected counts. All eleven targets there recover, including three genuine collisions without the order-three shortcut. The amendments preserve the original data and make no unbiased performance comparison. [pcc:unverified]



Unit: targets. Sources: orbits.json, residue-control.json, large-residue.json. Exact cases; no rates.

Actual constructor accounting. Candidate receipts record 1105 complete norm calls and 1105 changing-target constructors,195723 norm factor terms,630 canonicalizations and 439 target-independent G-graph points across exhaustive orbit qualification and walks. These include all control work; they are not an online timing interval. Public special tests, point arithmetic, modular roots, verification and reference/fresh replays are additional work. No candidate walk reads the reference scalar lookup or caches its full action table. The present norm backend remains linear in r, so an efficient first-variation evaluator is still needed for the primary curve. [pcc:unverified]

P-192 algebra qualification. The full-width checker verifies 12 manufactured collision controls from the declared 17G,31G,mu_G and relation_lambda_G fixtures: four linear,four quadratic and four projective-pole cases. Every accepted scalar is checked against its point; changed equations and proportional matrices reject. Eight public affine compositions on the existing unknown Q agree under independent affine/Jacobian point operations, and their tautological comparisons supply no scalar. Separately,-3 is a nonsquare in F_r and r=2mod 3, confirming the absence of the order-three path for P-192. No full P-192 norm, Mobius walk or unknown-scalar recovery is claimed by these controls. [pcc:unverified]

Verification and custody. Candidate and scalar-reference scientific outputs match for all five panels, with mode/cost receipts kept separately. Eight fresh-copy outputs reproduce exactly. A changed transporter and a changed full-width retained matrix reject with exit 101. Source remains native Rust with locked dependencies and source/amendment publications preceding affected checks. All five tautological attempts remain in the traces. The required legacy library check and actual errors are recorded in validation.json and failures.json. No cloud job or managed generator was started or modified. The algebraic proof is written, not a new Lean formalization or trusted PCC verdict. [pcc:unverified]

Next construction. The orbit method now retains the scalar equation and actual constructor costs. The next technical task is to test a compressed first-variation circuit under the actual horizontal CM relations, with target-dependent setup measured or bounded before assigning a useful orbit-search cost. The current implementation gives a replayable marked-action/canonicalization/recovery interface for that task. It does not infer a norm evaluator from a smaller class number or L-value. [pcc:unverified]

Sources checked. Round 60 norm proof and orbit-successor supply the marked primitive and proposed transporters. HAC chapter 3 Algorithm 3.34, https://cacr.uwaterloo.ca/hac/about/chap 3.pdf , supplies modular-root background; the matrix identities are derived explicitly. Prior-art searches did not establish novelty, and none is claimed for S 3, canonicalization or quadratic solving. KB tools/ wrapper are unavailable. ecdb tools and migrated wrapper/guide are unavailable; added synthetic curves are verified by native equation checks and complete prime-field enumeration. No registry fact or unqueried database status is substituted for those checks. [pcc:unverified]